

create_pg_std_cell_conn_pattern

NAME

`create_pg_std_cell_conn_pattern`
Creates a standard cell rail connection pattern. This pattern can be instantiated in a design by `set_pg_strategy` command to create standard cell rails.

SYNTAX

```
status create_pg_std_cell_conn_pattern
    pattern_name
    [-rail_width {top_width | @var1 bottom_width | @var2}]
    [-rail_shift {top_shift | @var1 bottom_shift | @var2}]
    [-rail_mask {top_mask | @var1 bottom_mask | @var2}]
    [-layers layers | @var]
    [-check_std_cell_drc true | false | @var]
    [-mark_as_follow_pin true | false | @var]
    [-parameters var_list]
```

Data Types

<code>pattern_name</code>	string
<code>top_width</code>	float
<code>var1</code>	string
<code>bottom_width</code>	float
<code>var2</code>	string
<code>top_shift</code>	string
<code>bottom_shift</code>	string
<code>layers</code>	list
<code>var</code>	string
<code>var_list</code>	list
<code>top_mask</code>	string (mask_one mask_two mask_three follow_pin)
<code>bottom_mask</code>	string (mask_one mask_two mask_three follow_pin)

ARGUMENTS

`pattern_name`
Specifies the name of the standard cell rail connection pattern.

`-rail_width {top_width | @var1 bottom_width | @var2}`
Specifies the top cell row rail width and bottom cell row width. This option is optional. By default, the tool uses the longest PG pin width. The units are defined in the technology file. Each width can also be specified by evaluating the `var1` and `var2` variables in `-parameters` option. The at character (@) is used as a prefix to evaluate the variable. If only one width is specified, it is applied to both top and bottom width values. If rail width is not specified, the standard cell pin width is used by default. This option is required when a design does not contain any standard cells.

`-rail_shift {top_shift | @var1 bottom_shift | @var2}`
Specifies the shift distance of rails. The units are defined in the technology file. The rail shift can also be specified by evaluating the `var1` and `var2` variables in the `-parameters` option. The at character (@) is used as a prefix to evaluate the variable. A positive distance shifts the bottom and top rails into the cell rows. A negative distance shifts the bottom and top rails out from the cell rows. If only one shift value is specified, it is applied to both top and bottom shift values. The shift value is applied to the center of rail with respect to the row boundary, that is, a shift value of 0 can align the center of rail with the row boundary. By default, the rails are aligned with standard cell pins.

`-rail_mask {top_mask | @var1 bottom_mask | @var2}`
Specifies the top cell row rail mask and bottom cell row mask. This option is optional. By default, the rail is not colored with any mask. Each mask can also be specified by evaluating the `var1` and `var2` variables in `-parameters` option. The at character (@) is used as a prefix to evaluate the variable. If only one mask is specified, it is applied to both top and bottom masks. The valid keyword for rail mask is one of `mask_one`, `mask_two`, `mask_three` and `follow_pin`. "follow_pin" refers to assigning same mask of std cell pins to std rails. Report command `report_pg_patterns` reports the rail mask in the same order, i.e. top and bottom masks.

`-layers layers | @var`

Specifies the list of layer names for standard cell connection. By default, the standard cell power ground pin layer is used for connection. The list of layers can be specified by evaluating the variable @var. The variable var is specified in -parameters option. The at character (@) is used as a prefix keyword to evaluate the variable. If layers are not specified, the standard cell pin layer is used for rail creation. This option is required when a design does not contain any standard cells.

- check_std_cell_drc true | false | @var
Specifies the option to check DRCs between the rail and the standard cells. For example, if standard cells are not placed, the DRCs between standard cell pins and PG rails can be ignored. This option can be set to "true", "false" or by evaluating the var variable in the -parameters option. The at character (@) is used as a prefix keyword to evaluate the variable. By default, the option is "false", and DRCs between rails and standard cells are ignored for faster runtime.
- mark_as_follow_pin true | false | @var
Specifies the output shape_use of the generated rails. If set the option false, the generated rail will have "lib_cell_pin_connect" attribute in shape_use. If it is true, it will be "follow_pin". The at character (@) is used as a prefix keyword to evaluate the variable. By default, the option is "false".
- parameters var_list
Specifies the list of parameters. Each parameter can be evaluated and used by other options in this command. This allows the pattern to be programmable and reused by passing different parameters in different situations. Each parameter var in the list var_list is a string and the at character (@) is used as a prefix to evaluate the parameter, such as @var. See the example section for more details.

DESCRIPTION

Creates a standard cell connection pattern. This pattern can be instantiated in a design by the set_pg_strategy command to create standard cell connections. For standard cell rails, macro and pad pins are not connected and floating segments are always kept.

EXAMPLES

The following example creates a standard cell connection pattern std_pattern_1 at layer M1.

```
prompt> create_pg_std_cell_conn_pattern \  
std_pattern_1 \  
-layers {M1}
```

The following example creates a standard cell connection pattern named std_pattern_2 at a programmable layer by using the parameter my_metal2. The rail width and shift distance are also specified by parameters, w and dist, respectively. When referring to this pattern with the set_pg_strategy command, you can program the actual layer name for the metal layer, for example "Metal2" or "M2". Parameters allow this pattern to be reused by designs with different technologies.

```
prompt> create_pg_std_cell_conn_pattern \  
std_pattern_2 \  
-parameters {my_metal2 w dist} \  
-layers {@my_metal2} \  
-rail_width {@w @w} \  
-rail_shift {@dist}  
prompt> set_pg_strategy \  
s1 \  
-core \  
-pattern {{name: std_pattern_2}{nets: VDD}{parameters: {M2 2 10}}} \  
-extension {{stop: 10} {layers: M2}}
```

MORE EXAMPLES

For more example, please start GUI and invoke the following command.
gui_show_task -task "Design Planning:PG Planning->Examples->Overview"

SEE ALSO

[compile_pg\(2\)](#)
[create_pg_composite_pattern\(2\)](#)
[create_pg_macro_conn_pattern\(2\)](#)
[create_pg_ring_pattern\(2\)](#)
[create_pg_wire_pattern\(2\)](#)
[remove_pg_patterns\(2\)](#)
[report_pg_patterns\(2\)](#)

[report_pg_strategies\(2\)](#)
[set_pg_strategy\(2\)](#)

Version V-2023.12-SP5-6
Copyright (c) 2025 Synopsys, Inc. All rights reserved.

Man(1) output converted with [man2html](#)